

EX. NO : 1**SIMPLE ML - PROGRAM – LINEAR REGRESSION****Date :14.07.2025**

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Aim:

To implement a simple linear regression model from scratch using Python, visualize the data points and regression line, and predict the value of y for a given value of x.

Procedure:**1. Import necessary libraries**

Import `matplotlib.pyplot` for plotting graphs.

2. Prepare the data

Use sample data points for x and y.

3. Calculate means

Calculate the mean of x values and mean of y values.

4. Calculate slope (m)

Use the formula for slope m:

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

5. Calculate intercept (c) Use the formula:

$$c = \bar{y} - m \cdot \bar{x}$$

6. Form the linear equation

Print the equation of the line $y = mx + c$

7. Predict y for a new value x=

Use the linear equation to predict y.

8. Plot the data points and regression line

- Plot original data points as scatter plot.

- Plot the regression line.
- Mark the predicted point on the graph.

9. Display the plot

Use `plt.show()` to visualize the results.

Program Coding:

```
import matplotlib.pyplot as plt

# Sample data
x = [1, 2, 3, 4, 5]
y = [2, 4, 5, 4, 5]

# Calculate means
mean_x = sum(x) / len(x)
mean_y = sum(y) / len(y)

# Calculate slope (m) and intercept (c)
numerator = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(len(x)))
denominator = sum((x[i] - mean_x)**2 for i in range(len(x)))
m = numerator / denominator
c = mean_y - m * mean_x

# Print equation
print(f"Equation of line: y = {m:.2f}x + {c:.2f}")

# Predict y for x = 6
x_test = 6
y_pred = m * x_test + c
```

```
print(f'Predicted y for x = {x_test} is {y_pred:.2f}')

# Plot original data

plt.scatter(x, y, color='blue', label='Original data')

# Plot regression line

regression_line = [m * xi + c for xi in x]

plt.plot(x, regression_line, color='red', label='Regression Line')

# Plot predicted point

plt.scatter(x_test, y_pred, color='green', marker='x', s=100, label=f'Prediction
(x={x_test})')

plt.title("Simple Linear Regression")

plt.xlabel("X")

plt.ylabel("Y")

plt.legend()

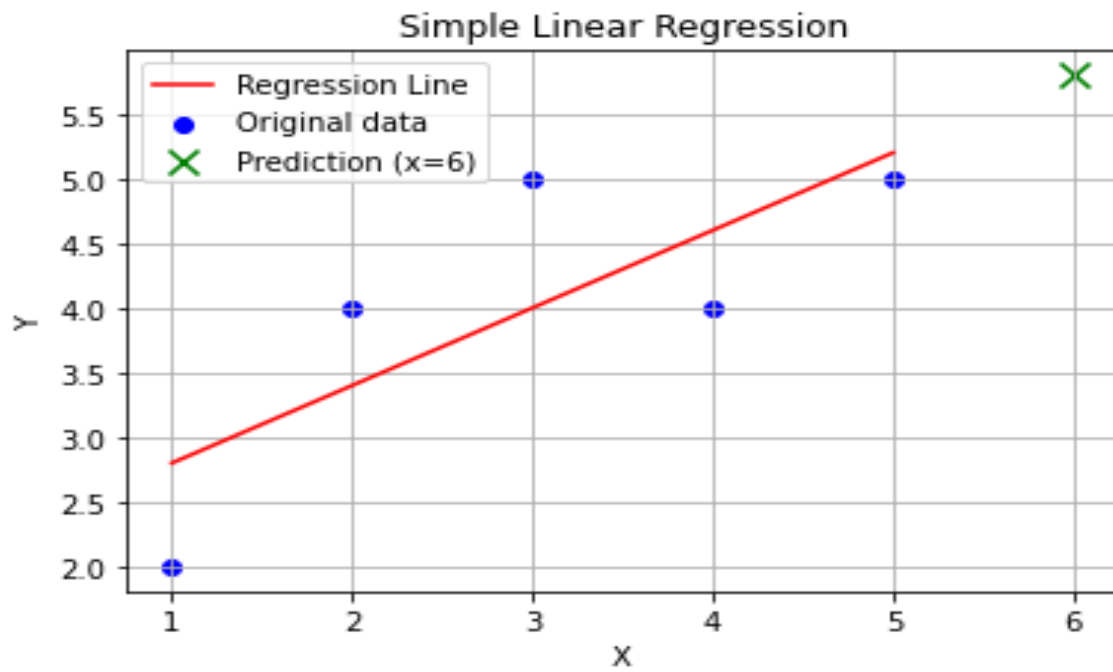
plt.grid(True)

plt.show()
```

Output:

Linear Regression Equation: $y = 0.60x + 2.20$

Predicted y for x = 6 is 5.80



Result:

Thus, the simple linear regression program was successfully implemented.